<u>Unit 4 Laser and Optical Fibre</u>

1. Discuss Interactions of radiation with matter
2. Set up an expression for energy density of radiation under equilibrium conditions in terms of Einstein Co- efficients
   OR Obtain the relation between Einstein Co-Efficients
3. Explain the necessary conditions to obtain laser beam (hint - high energy density, population inversion, metastable state)
4. Discuss the requisites for laser beam production (hint - active medium, pumping, optical cavity)
5. With neat diagrams explain the construction and working of semiconductor diode laser
6. Construction and working of optical fibre (hint - OF diagram and ray propagation diagram and its explanation, TIR explanation)
7. Derive an expression for acceptance angle in OF (hint - ray diagram and its explanation must)
8. Derive an expression for numerical aperture (or light gathering capacity) in an OF (hint - go up to condition for propagation $\sin \theta i < NA$)
9. Draw and explain refractive index profile and propagation modes for single mode step index, multi mode step index, graded index multimode (GRIN)
10. Discuss the different attenuation mechanisms in Optical fibre
11. Write the importance of single photon. Explain how a single photon is produced using attenuators.
12. What is the role of single photon in quantum computing? Mention different sources for single photon.
13. What are optical modulators? Mention different types of optical modulators.
14. What is Pockel's effect? Explain with a neat diagram.
15. Discuss Pockel's cell along with its principle, construction, working and applications.
16. What is Kerr effect? Describe with a suitable diagram.
17. Discuss Kerr cell along with its principle, construction, working and applications.
18. What are photo detectors? Give example.
19. Explain the principle, construction and working of Avalanche Photo Diode.
20. Explain the principle, construction and working of Single Photon Avalanche Diode.
21. With an appropriate diagram explain the construction and working of Superconductor Nanowire Single Photon Detector.
22. Discuss the construction and working of a Mach–Zehnder Interferometer (MZI) with a suitable diagram. Explain how interference is produced and mention its applications.
23. Numericals